

# Ben Bubnick

HEALTHCARE DATA STRATEGY · DATA GOVERNANCE · ANALYTICS DELIVERY

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## Summary

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Healthcare Data Consultant with 10+ years of experience leading the technical delivery and governance of analytics, data platforms, and ML solutions across clinical, operational, and revenue-cycle domains. Combines hands-on expertise in Python, SQL, Snowflake, SQL Server, AWS, Azure, and production data pipelines with leadership in data quality, source-to-target reconciliation, documentation, and shared standards. Skilled at turning ambiguous healthcare challenges into trusted analysis, reusable data models, real-time services, and stakeholder-ready reporting supported by testing, CI/CD, and observability. Known for aligning engineering, data science, operations, and business stakeholders while establishing transparent, scalable, and well-governed solutions for regulated environments.

## Experience

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### HHaEXchange

*Snowflake, dbt, SQL/Jinja, Python, Snowpark, pandas, Liquibase/GitHub Actions, FHIR/Aidbox*

HEALTHCARE DATA INTEGRATION CONSULTANT — CONTRACT VIA VEERSA TECHNOLOGIES

*Oct. 2025 - Jun. 2026*

Led healthcare source analysis, data-quality investigation and Snowflake modeling to produce trusted, standardized datasets for reporting and downstream use.

- Engineered unified and analytics Snowflake/dbt models spanning 55 tables across two organizations and two source-system configurations, reducing client onboarding time by 67%.
- Implemented reusable SQL/Jinja logic for lifecycle states, null handling, sentinel dates, and source-system variations across organization, office, caregiver, and patient domains, centralized 37% of post-source transformation logic in shared code
- Created or materially redesigned 30 SQL/dbt data-quality tests and designed macros to automate null, duplicate, referential-integrity, business-rule, lifecycle, and source-to-target checks.
- Profiled source schemas and converted ambiguous business rules into documented mappings, acceptance criteria, and onboarding standards, plus 12 data-quality runbooks that standardized investigation and onboarding practices.

### Adonis

*AWS: S3, Snowflake, SQLMesh, SQL, Python*

HEALTHCARE REVENUE CYCLE DATA CONSULTANT — CONTRACT VIA HEALTH DATA MOVERS

*Mar. 2025 - Jul. 2025*

Translated payer and revenue-cycle source data into standardized Snowflake models for repeatable client onboarding and reporting.

- Built, documented, validated, and maintained 54 SQLMesh models, supporting revenue-cycle analytics for a provider serving 4,598 families across 113 locations and processing 86,388 annual authorizations.
- Implemented automated validation across 54 SQLMesh models, including exact row-count audits, uniqueness checks, and not-null; achieved a 100% pass rate across 117+ configured audit applications at final delivery, including exact row-count reconciliation for 40 models, with no unresolved discrepancies.
- Operationalized 54 SQLMesh models with schema-change detection, audit logging, CI release gates, and reproducible recovery procedures.
- Authored implementation playbooks, plus internal handoff documentation and a QA runbook that established a repeatable healthcare-data validation workflow for an engineering team new to the domain.

### GrowData Analytics

*Ruby, RubyLLM, Tesseract OCR, GPT-4o mini, Claude Opus/Sonnet, RuboCop, RSpec*

INDEPENDENT DATA AUTOMATION & ANALYTICS CONSULTANT

*Sep. 2024 - Oct. 2025*

Built a document-to-data product that converted semi-structured invoices into validated, auditable transaction records.

- Built and validated a pre-production Ruby, OCR, and LLM invoice-automation MVP within 11 months, achieving 95% field-level accuracy across a stratified 500-invoice test suite (95% exact CI: 92.67%-96.79%).
- Instrumented the MVP with RSpec validation, audit logging, retry controls, error classification, and confidence-based routing, demonstrating 88% straight-through processing, 12% manual review, and under 1% complete system failure in testing.

## Arkos Health

*AWS: S3, Athena, Redshift, Glue, dbt, PySpark, Docker + AWS CLI, Tableau, Jupyter*

### SENIOR DATA ANALYST

*Jun. 2023 - Jan. 2025*

Delivered healthcare analytics, recurring reporting and data-quality investigations while improving the governed data foundation used by the BI team.

- Instrumented approximately 200 pipelines and scheduled jobs with monitoring, alerting, and 10+ failure classifications; authored three AWS/dbt incident runbooks that enabled failure detection within 5–15 minutes and typical recovery within 2–4 hours.
- Integrated Jupyter-based reporting and authored platform, architecture, and operating documentation, enabling six analysts to independently use the governed data lake for recurring and executive reporting.

## Merative

*Azure: Data Factory, Blob Storage, Functions, SQL DB, Synapse Analytics, Python, DevOps*

### SENIOR DATA SCIENTIST

*Jul. 2022 - Mar. 2023*

Led data validation, compliance and migration assurance for a regulated healthcare platform modernization.

- Engineered a seven-layer Python and SQL migration-acceptance suite covering partition-level reconciliation, schema compatibility, null handling, duplicates, field-level comparisons, aggregates, and business rules, achieving exact HDFS-to-Azure fidelity with zero unresolved discrepancies.
- Led HIPAA-compliant de-identification validation across demographic, encounter, and coded clinical data representing approximately 300 million patient lives, achieving a 100% pass rate before sign-off.

## IBM

*Hadoop/HDFS/CDH: Spark, Hive & Impala, Kafka, Python, JRuby, SQL, Pig, Docker + Gradle*

### SENIOR DATA SCIENTIST

*Nov. 2021 - Jul. 2022*

Delivered client analytics and large-scale migration validation for healthcare administrative data.

- Developed an MVP surgical-scheduling optimizer and a CNN/RNN case-card prediction proof of concept, automating case-card generation and resource allocation across operating rooms, time slots, and personnel.
- Engineered a reusable Spark, Hive, and Impala analytics framework powering 50+ downstream prior-authorization lag-time and approval-bottleneck reports, with daily operational execution and monthly or quarterly regulatory and financial snapshots.

### SENIOR DATA ARCHITECT

*Dec. 2020 - Nov. 2021*

Led clinical-data integration and unstructured-text analysis at enterprise scale.

- Architected Spark and Hadoop pipelines that integrated and normalized more than 100 billion structured and semi-structured clinical records, supporting four EPM applications, 30+ risk models, and hundreds of standardized measures and registries.
- Replaced static mappings with executable, metadata-driven instruction sets adopted across approximately 200 source integrations, standardizing EDI batch schemas while enabling modular integration templates, automated Python/Jupyter lineage tracing, and end-to-end validation gates.

### DATA ARCHITECT

*Nov. 2017 - Dec. 2020*

Owned client-facing integration, discrepancy resolution and data-quality improvement across multi-organization healthcare networks.

- Designed standardized SQL validation frameworks and client-review gates across approximately 100 client integrations, reducing Jira-tracked validation rework hours by 25% and mapping rework hours by 45% across initial delivery and subsequent corrections.
- Led architecture and delivery of a multi-organization healthcare integration spanning 70+ heterogeneous EHR sources and isolated organizational schemas; implemented provider-scoped patient filtering to enforce data-sharing rules, and coordinated with engineering teams.

### SOFTWARE DEVELOPER, DATA PLATFORM

*Sep. 2016 - Nov. 2017*

Built internal data-curation and knowledge-transfer tools that reduced costs and onboarding effort.

- Engineered an in-house HL7 v2/v3 parsing pipeline in Pig, supporting approximately 30 internal users, replacing the Mirth-based workflow, and eliminating \$30,000 in annual vendor costs.
- Built a Python, Ruby, and SQL extraction tool that reduced total client-integration effort by 28%, measured through eliminated story points and corroborated by lower average engineering hours per implementation.

## DATA SCIENTIST

Nov. 2015 - Sep. 2016

Coordinated client integration delivery, contractor work and process automation.

- Directed technical delivery across 36 client data integrations, coordinating 12 remote contractors and establishing code-review, QA, and issue-resolution standards.
- Authored 125+ technical knowledge-base articles, approximately 30% of the team's published content, to standardize integration development and troubleshooting practices.

## Explorys

*Hadoop/HDFS: Ruby, Pig, React, NodeJS, JavaScript, SQL, Oracle, Jenkins*

## DATA SCIENCE CONSULTANT — CONTRACT VIA TEKSYSTEMS

*Jun. 2015 - Nov. 2015*

Supported clinical-data integration and population analytics across more than 10 client data warehouses.

- Diagnosed Hadoop and Oracle production-ingestion failures and implemented Jenkins-backed scheduling and recovery controls across 10+ client data warehouses.

## Other Experience

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SOFTWARE ENGINEER, **AmTrust Financial Services**, *Mar. 2014 - Jun. 2015.*

COLOR SCIENTIST, **PPG Industries**, *Apr. 2012 - Jun. 2013.*

ADJUNCT PROFESSOR, **Multiple Institutions**, *Aug. 2010 - May. 2019.*

## Volunteer Experience

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### HSEA

#### ETL BRIGADE

*Feb. 2025 - May. 2025*

Built a reusable PostgreSQL data model to help community health organizations prototype analytics tools for claims, visits, and eligibility.

### CSforAll

#### INSTRUCTOR

*Oct. 2018 - May. 2019*

Our goal was to provide computer science instruction in all of the District's high schools, with a special emphasis on students who do not have access to a computer or a stable Internet connection at home.

### HIT in the CLE

#### COACH

*Mar. 2017 - Mar. 2019*

We were using a data science competition to educate students understand the potential of a career in health information technology and to assist IT companies in Northeast Ohio in developing a talent pipeline.

### Code for America

#### ETL GUILD

*Jan. 2015 - Jun. 2015*

The initiative was brought about to apply data science tools to improve and expand public record access.

## Professional Recognition

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**HDM**, SPECIAL RECOGNITION (2025), Recognized for exceptional performance and commitment in client delivery.

**Arkos**, ARKOS HEALTH BI STAR (2023), Recognition for Achieving Expeditioner Rank, Salesforce Trailhead.

**Merative**, MANAGER RECOGNITION FOR TRANSITION LEADERSHIP (2022), Recognized for systems migration efforts.

**IBM**, GOLD CHAMPION (2018 - 2022), Recognized for 1,477 classroom hours delivered and 43 certificates earned.

**IBM**, LAB SERVICES AWARD (2018), Recognized for coordinating internal teams to deliver solutions on time.

**IBM**, EMINENCE AND EXCELLENCE AWARD (2017), Recognized for developing a new data curation tool.

**PPG**, COLOR LAB AUTOMATION AWARD (2012), Recognized for expanding data collection and analysis reporting.

## Speaking Engagements

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### AI Hackathon 2020: Surgery Scheduling Optimization

*"An algorithm uses past utilization and current waitlist info to predict surgery duration.*

Dec. 2020

IBM

### REACH Initiative: IBM Watson Care Insights

*Analyze patient records and real-time data via HL7 feeds, integrate insights into physician's EHR workflow.*

2019

IBM

### Content & Data Analytics Learning Exchange

*Introduction of implementation of machine learning algorithms using real-world health data analytics.*

2018

IBM

### 2018 Innovation Research Interchange Fall Networks Conference

*Using NLP to extract insight from unstructured data and transform it from a data lake.*

Sep. 2018

IRI

### Cross Team Training Series

*A 7-part series on using Hadoop for data ingestion that shows advantages of data lakes.*

2016

EXPLORYS

## Publications

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### Reproducible Postgres load of Tuva seed datasets

*Reproducible Postgres load of Tuva seed datasets with robust validation.*

Aug. 2025

HSEA

### Surgical Scheduling Optimization and Procedure Duration Prediction <sup>a</sup>

*Improvement on traditional fixed ratio methods for total procedure time estimation.*

Dec. 2020

IBM

### Process and Requirements for CCD Data Integration

*Processes and requirements for overcoming the many difficulties with CCD data curation.*

Oct. 2017

IBM

### AI Paint Formula Prediction from Spectroscopic Measurements <sup>b</sup>

*Naive bayesian color and effects matching algorithm of complex automotive paint formula.*

May. 2015

SHINYAPPS.IO

### Characterization of Surface Coatings using RT Models for Color Matching

*Approximation of Chandrasekhar RT model for atmospheric scattering applied to color matching.*

Feb. 2013

USPTO

### Massive Stellar Clusters in the Disk of the Milky Way Galaxy <sup>c</sup>

*Spectroscopic and Photometric analysis of two open clusters in the galactic plane.*

Dec. 2010

OHIOLINK ETD

### VizieR Online Data Catalog: JHK photometry in Cluster [BDS2003] 107

*Spectra were obtained 2005 June 1415 at the 3.0 m NASA Infrared Telescope Facility (IRTF) on Mauna Kea, Hawaii, using SpeX.*

May. 2008

VIZIER

### Near-Infrared Spectroscopy of Candidates Members of the Galactic Cluster [BDS2003] 107

*New near-infrared classification spectra for nine candidate members of the galactic cluster [BDS2003] 107 (Cluster 107).*

Feb. 2008

PASP

### A Novel Method for Low-Toxic Photo-Etch Printing <sup>d</sup>

*Application of aquatint etching techniques to photo-etch procedures.*

May. 2003

MAPC

## Education & Credentials

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2022 **IBM Data Science Professional Certificate** <sup>a</sup>

IBM

2015 **Certificate of Specialization** Data Science <sup>b</sup>

Johns Hopkins University

2010 **Master of Science** Astrophysics <sup>c</sup>

University of Cincinnati

2004 **Bachelor of Fine Arts** Printmaking <sup>d</sup>

Ohio University